



The Digestive System

Lecture 1

Overview of the Digestive System

- **General concepts**
- Human GI tract
- Mobility of the digestive system

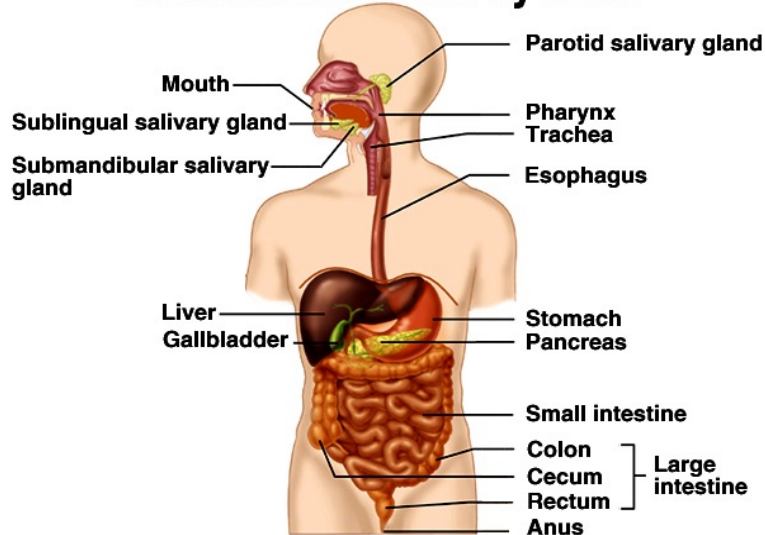


Digestive System

The alimentary **canal** and **accessory** digestive organs as an integrated system responsible for the ingestion, digestion, and absorption of food.

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Gastrointestinal system



Two Types of Digestion:

- 1) Mechanical:
Solid food masses are torn apart, ground up and vigorously mixed with digestive secretions to prepares food for chemical digestion. e.g.:
Chewing
Mixing of food with saliva by the tongue
Churning food in the stomach
Segmentation (rhythmic local constrictions of the intestine)
- 2) Chemical:
Complex food molecules are broken down to their chemical building blocks by enzymes secreted by various digestive glands.

Classes of Nutrients

- 1) Protein:
broken down into peptides->amino acids, which are absorbed by the small intestine, and transported into the blood for protein building and as an energy source.
- 2) Carbohydrates:
broken down into oligosaccharides->monosaccharides (glucose, fructose, & galactose) which are absorbed into the bloodstream.
- 3) Fats:
mainly triglycerides in the diet. Broken down into glycerol and free fatty acids and then absorbed by the intestine.

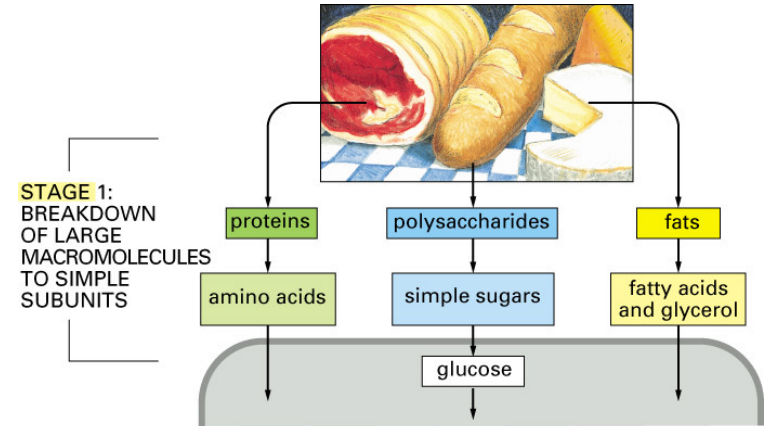


Figure 2-70 part 1 of 2. Molecular Biology of the Cell, 4th Edition.



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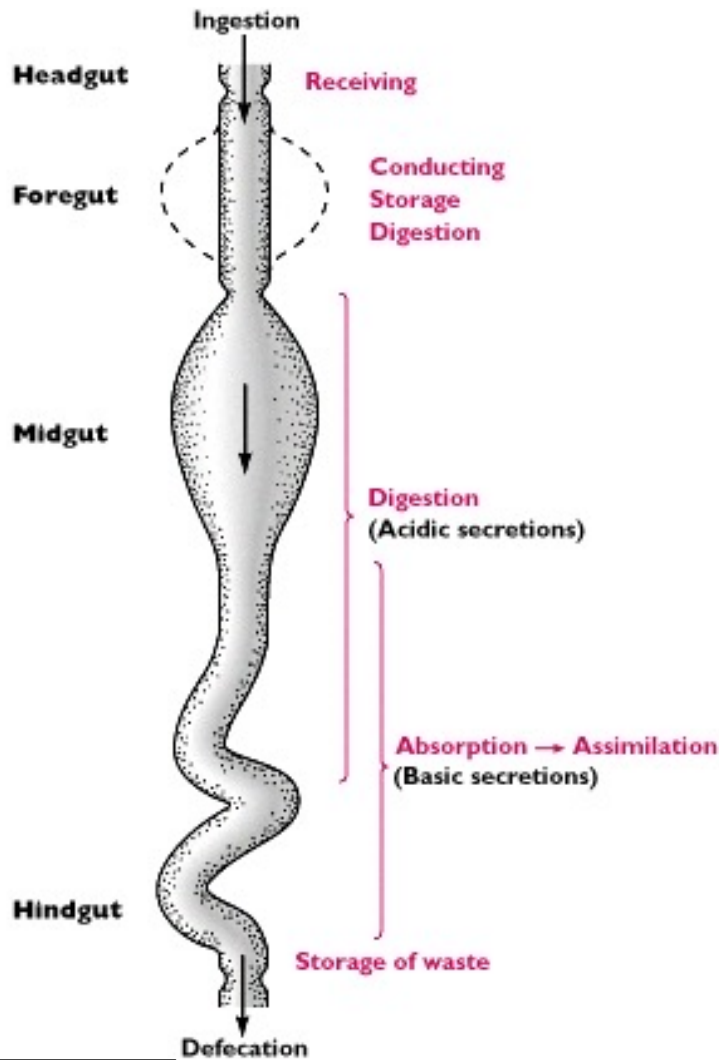
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- Mobility of the digestive system



The plan of a generalized animal GI tract



- Receiving
- Conducting, storing
- Digestion, absorption
- Absorbing water, defecating

1) Headgut: food intake & primary processing

Mouth (oral or buccal cavity):

Chewing (mechanical digestion)

Saliva from salivary glands

- 1) helps mechanical digestion by lubricating food &
- 2) contains enzymes for chemical digestion of starchy food.

pharynx:

Propel food into the esophagus (swallowing).

2) Foregut: food conduction, storage, & digestion

Esophagus:

Conduct food from the mouth to the stomach.

Stomach:

A reservoir to receive food all at once

Mechanical mixing of food with saliva and stomach secretion

Partial digestion

No absorption

3) Midgut: digestion & absorption

Small intestine:

Mechanical shaking and mixing of food

Absorption for all food products

Pancreas/liver:

Hydrolytic enzymes from both of them for chemical digestion

Bile from the liver facilitates fat digestion

4) Hindgut: absorption of water & ions, & excretion

Large intestine (colon):

Accumulation & dehydration of digestion waste

Absorption of water, ions, & some vitamins

Rectum, anus:

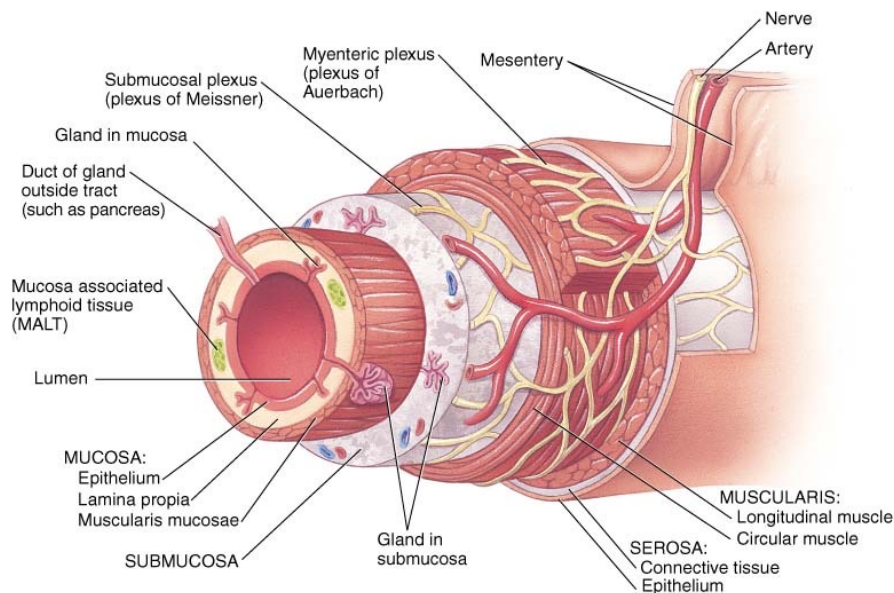
Expulsion of feces from the GI tract

Gasterintestinal organs



Organ	Exocrine secretions	Functions
Mouth and pharynx Salivary glands	Salt and water Mucus Amylase	Chewing begins; initiation of swallowing reflex Moisten food Lubrication Polysaccharide-digesting enzyme
Esophagus	Mucus	Move food to stomach by peristaltic waves Lubrication
Stomach	HCl Pepsin Mucus	Store, mix, dissolve, and continue digestion of food; regulate emptying of dissolved food into small intestine Solubilization of food particles; kill microbes Protein-digesting enzyme Lubricate and protect epithelial surface
Pancreas	Enzymes Bicarbonate	Secretion of enzymes and bicarbonate; also has nondigestive endocrine functions Digest carbohydrates, fats, proteins, and nucleic acids Neutralize HCl entering small intestine from stomach
Liver	Bile salts Bicarbonate Organic waste products and trace metals	Secretion of bile; many other nondigestive functions Solubilize water-insoluble fats Neutralize HCl entering small intestine from stomach Elimination in feces
Gallbladder		Store and concentrate bile between meals
Small intestine	Enzymes Salt and water Mucus	Digestion and absorption of most substances; mixing and propulsion of contents Food digestion Maintain fluidity of luminal contents Lubrication
Large intestine	Mucus	Storage and concentration of undigested matter; absorption of salt and water; mixing and propulsion of contents; defecation Lubrication

Microscopic Structure of the GI tract



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- Mucosa – mucous membrane (epithelial, connective, smooth muscle tissues)
- Submucosa – glands/loose connective tissue
- Muscularis – longitudinal and circular fibers
- Serosa – visceral peritoneum



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- **Mobility of the digestive system**

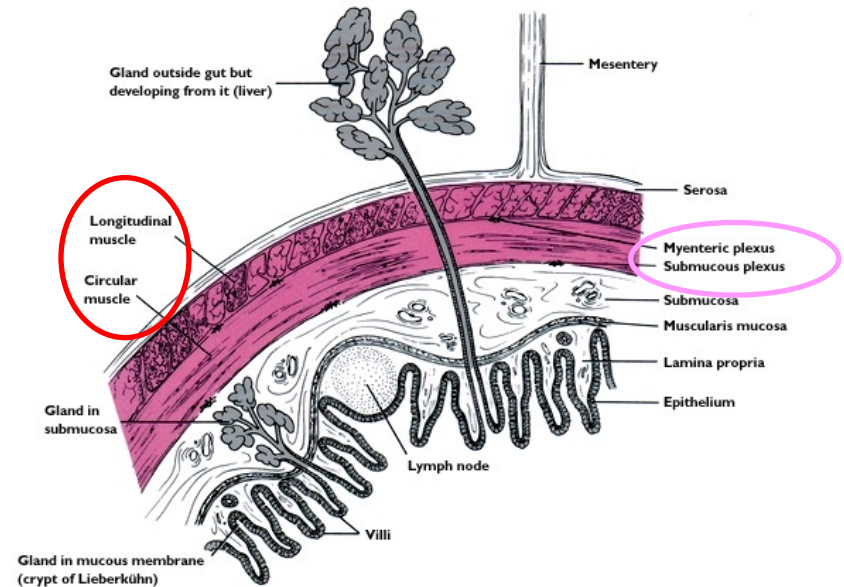


Functions of GI tract's Movement

1. Translocation of food from point A to B.
2. **Mechanical breakdown of food.**
3. Mix food for continual renewal of material in contact with the secreting & absorbing surfaces of the GI tract.

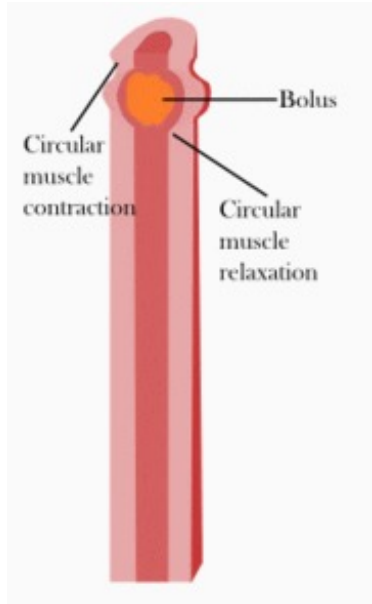


Food moves down in the GI track in the absence of gravity.



GI mobility needs involvement of muscles

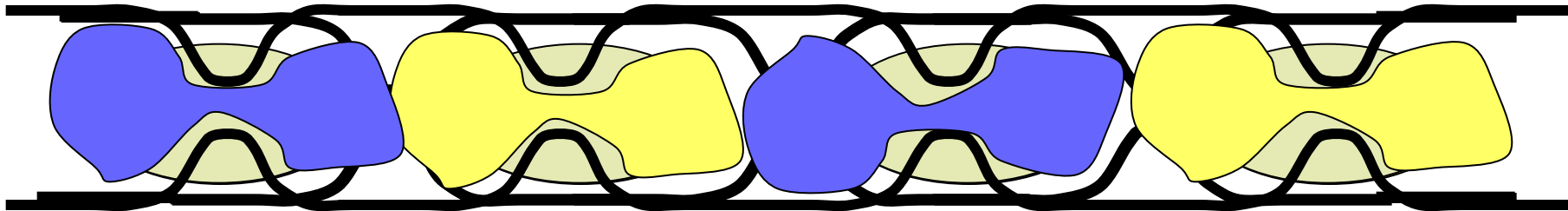
Types of GI Motility



- 1) **Peristalsis**- traveling wave of constriction caused by action of longitudinal and circular smooth muscles.
 - Usually in an aboral direction. Initiation mainly results from distention of the gut wall.
 - Main function is propulsion.
 - Site: esophagus
distal stomach
small intestine
large intestine



- 2) **Segmentation** -asynchronous constriction of gut tube; the longitudinal muscle NOT involved
 - Breaks down large food particles and mixing.
 - Site: small and large intestines



Control of Motility

1) Intrinsic control

Mediated by enteric nervous system. This type of control is modulated by stretching of the GI tract and substances in the chyme (partly digested food in the stomach).

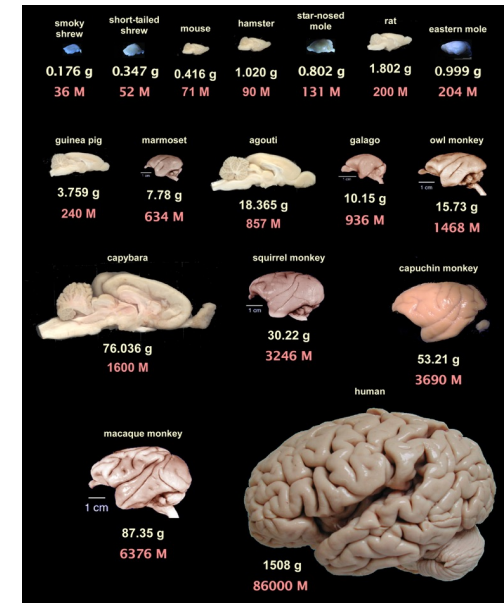
2) Extrinsic control

a. GI peptide hormones secreted by the GI tract in response to chemical stimulants

b. Sympathetic (inhibitory) & parasympathetic (excitatory) nervous systems

Enteric nervous system

“little brain in the gut”



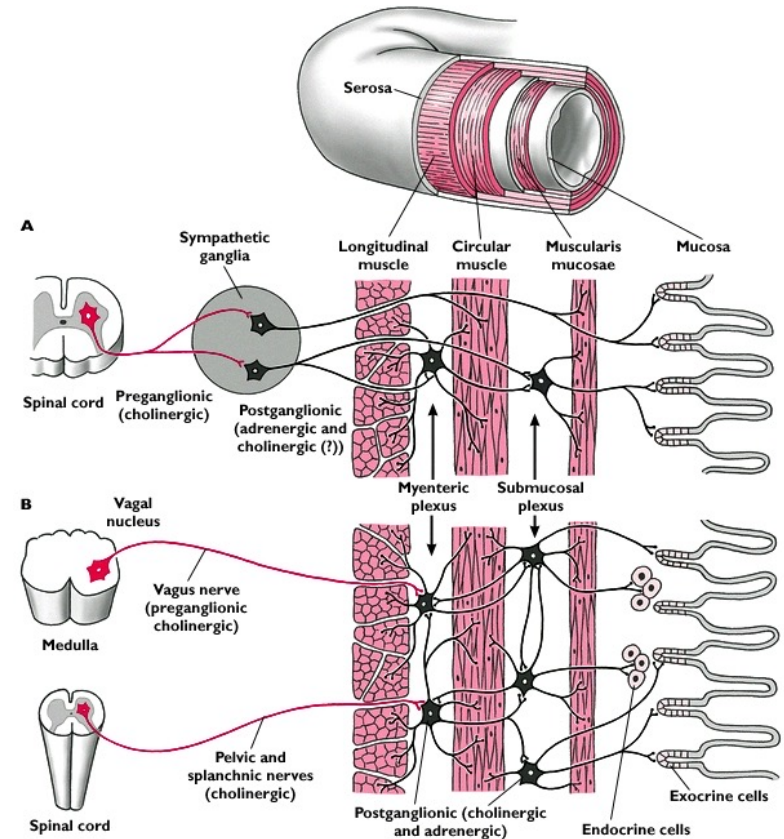
- consist of myenteric and submucosal plexus.
- contains 100 million neurons (more than a mouse brain) , including sensory neurons, interneurons, motor neurons.
- Organized as an independent integrative system.
- Plexus: a network of neurons



Innervation of the GI Tract

Sympathetic Nervous System

Parasympathetic Nervous System



Hirschsprung's disease

In children with Hirschsprung's disease, nerves fail to form in all or part of the large intestine (colon). Waste from digestion cannot pass through the part of the colon lacking nerve tissue. The normal colon swells with blocked stool.

Other Inputs

Central Nervous System

Sensory neurons:

Mechanoreceptors

Chemoreceptors

Thermoreceptors

Interneurons:

Reflexes

Motor programs

Motor neurons:

Inhibitory

Excitatory

Muscle

Secretory epithelium

Endocrine cells

Blood vessels

Enteric Nervous System